LOW-RESOURCE ASR WITH PRETRAINED SPEECH SELF-SUPERVISED REPRESENTATIONS

Yuhua Luo *Haomin Hu* *Yuecheng Han*

Course Project
Yuhua Luo: 524531910035 Haomin Hu: 524531910040 Yuecheng Han: 524531910029

ABSTRACT

This project studies low-resource automatic speech recognition (ASR) with pretrained speech self-supervised learning (SSL) representations. We implement a LibriSpeech pipeline with transcript normalization, JSONL manifests, character-level CTC training, greedy decoding, WER/CER evaluation, and qualitative error analysis. Using approximately one hour from LibriSpeech `train-clean-100`, we compare a small log-mel CTC baseline with wav2vec 2.0 and HuBERT encoders followed by a CTC head. A focused layer ablation shows that an intermediate wav2vec 2.0 layer is much stronger than the final layer under the frozen-encoder setting. The best system is HuBERT base with a frozen encoder, reaching 0.662 WER and 0.236 CER on the evaluated `test-clean` set.

Index Terms— speech recognition, self-supervised learning, wav2vec 2.0, HuBERT, CTC

1. INTRODUCTION

Low-resource ASR is difficult because the model must learn an acoustic-to-text mapping from limited paired speech and transcripts. Speech SSL pretraining addresses this limitation by learning robust acoustic representations from large unlabeled audio corpora before downstream supervised training. This project follows that approach and builds an English ASR system centered on pretrained wav2vec 2.0 [1] and HuBERT [2] representations.

The goal is not to compare a large model zoo. Instead, the study asks whether pretrained SSL hidden states help in a one-GPU, low-resource LibriSpeech setting, and whether representation choice matters when the downstream model is deliberately small. The implemented system therefore includes one conventional baseline, two SSL encoders, and one single-factor layer ablation. We focus on continuous hidden-state representations rather than discrete units; this choice keeps the downstream system simple enough to train and evaluate reliably within the course-project compute budget.

2. METHOD

2.1. Data and Normalization

The experiments use LibriSpeech [3]. To avoid unreliable long tar downloads, the data preparation script reads the `openslr/librispeech_asr` parquet mirror, materializes FLAC files, and writes LibriSpeech-style JSONL manifests. The training split contains 269 utterances, totaling 1.004 hours, sampled from `train-clean-100`. The development and test manifests contain 2703 and 2620 utterances from `dev-clean` and `test-clean`.

Transcripts are lowercased, punctuation is removed, apostrophes are preserved, and repeated whitespace is collapsed. The CTC vocabulary is character-level and includes a blank symbol, a padding symbol, an unknown symbol, letters, apostrophe, and space.

2.2. Log-Mel CTC Baseline

The baseline extracts 80-dimensional log-mel features with torchaudio and feeds them to a small feed-forward encoder with four hidden layers. A linear CTC head predicts character labels. This model has no pretrained speech knowledge and serves as the low-resource non-SSL comparison.

2.3. SSL CTC Models

The SSL systems use Hugging Face Transformers encoders followed by a small CTC projection head. We evaluate wav2vec 2.0 base and HuBERT base. In the main SSL runs, the encoder is frozen and only the CTC head is trained. This keeps the experiment feasible on one 8 GB GPU and isolates the usefulness of pretrained hidden states.

For layer analysis, we compare wav2vec 2.0 final-layer features against wav2vec 2.0 layer 6 features while keeping the dataset, optimizer, frozen encoder setting, CTC head, and decoding procedure unchanged.

2.4. Experimental Controls

All systems use the same transcript normalization, character vocabulary construction, CTC objective, and greedy CTC

Table 1. Low-resource ASR performance using 1 hour of LibriSpeech train-clean-100.

System	Dev WER	Test WER	Test CER
HuBERT last	0.656	0.662	0.236
Log-mel	1.000	1.000	1.000
wav2vec2 last	0.986	0.985	0.596
wav2vec2 layer 6	0.659	0.667	0.262

decoding. No external language model is used. This makes the comparison conservative: language-model fusion could improve absolute WER, but it would also obscure how much information the acoustic representation itself contributes. The log-mel system is the non-pretrained control, the HuBERT and wav2vec 2.0 systems test the effect of SSL pretraining, and the wav2vec 2.0 layer-6 run isolates the hidden-layer choice while leaving the rest of the pipeline fixed.

3. EXPERIMENTAL SETUP

All completed training used PyTorch 2.7.1 with CUDA 12.8 on an NVIDIA GeForce RTX 4060 Laptop GPU. The log-mel baseline was trained for 8 epochs. The SSL systems were trained for 5 epochs with batch size 2 and AdamW. Decoding uses greedy CTC collapse without an external language model.

WER and CER are the primary metrics. Real-time factor (RTF) is also recorded. For SSL models, the final retained evaluation uses a 30-second maximum utterance duration, covering 2694/2703 development utterances and 2611/2620 test utterances. A strict 60-second re-evaluation was attempted for the full split but the wav2vec 2.0 run became unexpectedly slow, so it was stopped and the high-coverage evaluation was retained. The log-mel run has WER/CER of 1.000 under both limits. The reported results should therefore be interpreted as a high-coverage low-resource comparison rather than as a full LibriSpeech benchmark submission.

4. RESULTS

The log-mel baseline collapses to empty or near-empty hypotheses, producing 1.000 WER and CER. The final layer of frozen wav2vec 2.0 is better at the character level, but its word-level performance remains weak. The layer ablation is important: changing only the wav2vec 2.0 representation from the final layer to layer 6 improves test WER from 0.985 to 0.667 and test CER from 0.596 to 0.262. HuBERT base is the strongest main SSL model, reaching 0.662 test WER and 0.236 test CER.

These results support the central claim that pretrained speech SSL representations are useful in low-resource ASR, but they also show that the choice of hidden layer can matter as much as the choice between SSL model families under frozen-encoder CTC training. The result is also a useful negative finding: using a pretrained encoder does not automatically solve the task. The final wav2vec 2.0 layer remains close to the failing baseline in WER, which suggests that frozen final-layer representations may be poorly matched to a very small character CTC head without fine-tuning or an auxiliary language model.

5. ERROR ANALYSIS

The worst log-mel errors are deletions: the model often emits an empty hypothesis for complete reference sentences. This indicates that the small baseline learns the CTC blank-dominant solution under one hour of supervision.

The SSL systems make more interpretable errors. Common HuBERT failure cases include substitutions on names or rare words, such as “stephanos dedalos” being recognized as a phonetically similar but misspelled phrase, and deletions in short utterances such as “squeak squeak”. Many errors preserve some phonetic structure but miss exact spelling and word boundaries. The wav2vec 2.0 layer-6 system shows similar substitutions and deletions, while the final-layer wav2vec 2.0 system has substantially more broad word-level errors. The pattern is consistent with a model that has learned useful phonetic structure but still lacks robust lexical and word-boundary modeling. This is expected for greedy character CTC decoding without a language model.

6. LIMITATIONS

The study is intentionally scoped to continuous SSL hidden states. It does not measure discrete-unit token rate, bitrate, or codebook size, because no quantizer is part of the implemented ASR system. The comparison is also limited to frozen encoders and one hour of labeled training data. Fine-tuning the SSL encoder, adding a word-piece decoder, or using an external language model would be natural next steps, but those changes would test different hypotheses from the representation-focused comparison reported here.

7. REPRODUCIBILITY

The submission includes the source code, configs, generated manifests, metrics, prediction CSV files, error examples, and report source. The main commands are listed in `README.md` and `REPRODUCIBILITY.md`. The final experiment plan is `configs/experiment_plan.yaml`. The measured metrics are stored under the `outputs` directory.

8. CONCLUSION

This project implements a complete low-resource SSL ASR pipeline. It satisfies the ASR task requirement, uses pretrained SSL representations, reports WER and CER, includes a conventional baseline, compares wav2vec 2.0 and HuBERT, and adds a clean hidden-layer ablation. The strongest result comes from HuBERT base, while the wav2vec 2.0 layer ablation shows that intermediate representations can be much more useful than the final layer for frozen CTC training with only one hour of labeled speech.

9. REFERENCES

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